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DATA STRUCTURES
ALGORITHMS
COMPUTER NETWORKS
DATABASE SYSTEMS

Major Course (Elective-2): Artificial Intelligence

Level	Semester	Course Code	Course Name	Credits	Teaching Hours	Exam Duration	Max Marks
5.5	VI	**112225**	**Artificial Intelligence**	2	30	2 Hrs.	30

Course Objectives:	1. Apply knowledge of Artificial Intelligence to solve complex engineering problems. 2. Identify the problem, formulate, and analyze AI-based problems using search algorithms. 3. Design intelligent systems and AI models using suitable algorithms for problem-solving, planning, reasoning, and decision-making applications. 4. Enable students to understand the foundational principles, structural components, and practical utility of expert systems in mimicking human decision-making. 5. To enable students to deconstruct human communication and identify the mathematical boundaries that separate natural human languages from formal programming languages.			
Course Outcomes:	Students will be able to - 1. Explain the concepts, applications, and problem-solving approaches of Artificial Intelligence. 2. Apply blind search techniques for AI problem solving. 3. Implement heuristic search methods including Hill Climbing, Best First Search, A*, AO*, Generate and Test, and Means-End Analysis. 4. To architect, engineering, and deploy functional, knowledge-based systems that capture human expertise and deliver automated, rule-based decision-making across enterprise and web environments. 5. To architect, audit, and debug computational pipelines that convert highly ambiguous human speech or text into structured, machine-interpretable logical data.			
Unit System	**Contents**	**Workload Allotted**	**Weightage of Marks Allotted**	**Incorporation of Pedagogies**
Unit I	**Introduction**: Introduction to artificial intelligence, AI problems and Techniques background, applications, Turing test, Artificial General Intelligence, Super AI, rational agent approaches to AI, introduction to intelligent agents, their structure, behavior and task environment.	7 Hrs.	7 Marks	Use any pedagogical technique suitable for the topic
Unit II	**Problem Solving and Searching Techniques:** Problem characteristics, production systems, control strategies, breadth-first search, depth-first search, hill climbing and its variations, heuristics search techniques: best-first search, A* algorithm, constraint	8Hrs.	8 Marks	

	satisfaction problem, means-end analysis, min-max and alpha-beta pruning algorithms.			
Unit III	**Expert systems:** Introduction, basic concepts, structure of expert systems, the human element in expert systems how expert systems works, problem areas addressed by expert systems, types of expert systems, expert systems and the internet interacts web, knowledge engineering, scope of knowledge, difficulties, in knowledge acquisition methods of knowledge acquisition, machine learning, intelligent agents.	8 Hrs.	8 Marks	
Unit IV	**Understanding Natural Languages:** Components and steps of communication, the contrast between formal and natural languages in the context of grammar, Chomsky hierarchy of grammars, parsing, and semantics, Parsing Techniques, Context-Free and Transformational Grammars, Recursive transition nets.	7 Hrs.	7 Marks	
References:	**Course Material/Learning Resources** **Text books:** 1. Russell, Stuart, J. and Norvig, Peter, *Artificial Intelligence – AModern Approach*, Pearson, 4th edition, 2020.. 2.Patterson, DAN,W, *Introduction to A.I. and Expert Systems* – PHI, 2007 **Reference Books:** 1. Artificial Intelligence Elaine Rich and Kevin Knight 2. Tata McGraw Hill, 2nd Edition 3. Computational Intelligence Eberhart Elsevier Publication 4. Artificial Intelligence: A New Synthesis Nilsson Elsevier Publication 5. Artificial Intelligence with Python Prateek Joshi Packt Publishing Ltd 6. Artificial Intelligence Saroj Kausik Cengage Learning **Weblink to Equivalent MOOC on SWAYAM if relevant:** https://onlinecourses.swayam2.ac.in/e-learning/preview/cec21_cs08 https://onlinecourses.nptel.ac.in/e-learning/preview/noc21_cs42 https://onlinecourses.swayam2.ac.in/e-learning/preview/cec26_cs04 **Weblink to Equivalent Virtual Lab if relevant:** https://ai1-iiith.vlabs.ac.in/ **Any pertinent media (recorded lectures, YouTube, etc.) if relevant:** https://www.youtube.com/watch?v=V5zSFToG7eg https://www.youtube.com/watch?v=O-FZ4Q8RXds https://www.youtube.com/watch?v=Ur3NZZo9hEI			

Assessment Rubric: Internal for Theory Course (Major Elective-2)

SANT GADGE BABA AMRAVATI UNIVERSITY, AMRAVATI THEORY EXAMINATION B.Sc. III, SEMESTER – VI (NEP)			
Theory	**Course Code:** 112225	**Course Title: Artificial Intelligence**	**Max Marks: 20**
Sr. No.	**Assessment Criteria**		**Marks**
1	Any Two of the Activity (weekly or monthly assignment/ Unit Test/Class Test) **(CAT-1)**		**10**
2	Any One of the Activity includes Open Book Test, MCQ Tests, / Theory related Class activity/Task etc.) etc. **(CAT-2)**		**5**
3	Participation in any one Activity includes Quiz, Seminar, Group Discussion, Industry visit, Case Study, Mini project, innovative work, etc. based on Theory Course. Involvement in subject related activities such as Conference, Workshop, Coding Competitions etc. **(CAT-3)**		**5**